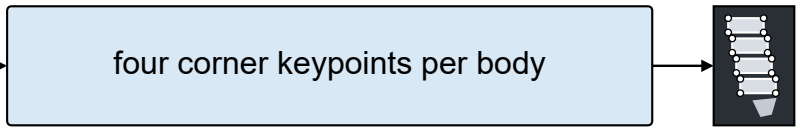
a **Detectors** neither trained on the test C-arm; neither names a level

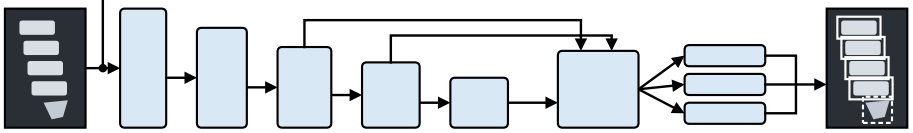
YOLO11m-pose keypoint detector, trained on films



body → b, c

no sacrum box

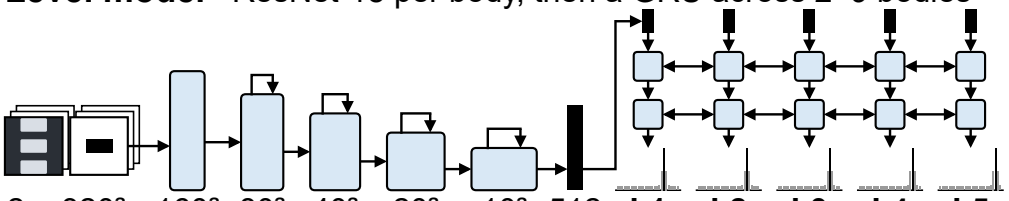
YOLO11m render-trained, fine-tuned on real frames of two other devices



vertebra

sacrum → c

b **Level model** ResNet-18 per body, then a GRU across 2–9 bodies



one 512-vector

GRU, two layers, both directions

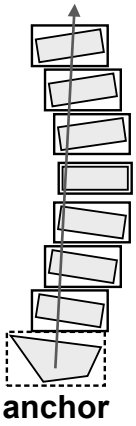
posterior

27 levels



C1–C7 T1–T13 L1–L6 sacrum

c **Count**



from the detected sacrum; no names

d **Coupling** four alphabets name the ordinals

posterior	ordinary	L6	T13	both
	T11	T12	T12	**T13**
	T12	L1	**T13**	L1
	L1	L2	L1	L2
	L2	L3	L2	L3
	L3	L4	L3	L4
	L4	L5	L4	L5
	L5	**L6**	L5	**L6**
	sacrum	sacrum	sacrum	sacrum
		call		

score: mean log posterior of an alphabet's names
call: the alphabet that scores highest
stop: a classifier name differs from the call, or a low margin over the next-best alphabet